Governance Under Adversarial Strategy Injection in Sequential Commons

Ameer Alhashemi

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Abstract

We study governance under adversarial strategy injection in sequential commons through two linked studies. The first study uses a minimal Fishery Commons setting to compare central governance signals under repeated strategy turnover and held-out regime shifts. A matched mutation-versus-live-LLM baseline shows that monitoring with sanctions reduces held-out collapse relative to no governance for both injector types, a broader top-down sweep over partner mixes and pressure levels identifies adaptive quotas as the strongest overall signal, and a learned-policy self-play validation in the medium tier recovers the same ordering: adaptive quotas beat no governance on held-out collapse, stock, and welfare while also outperforming the other governed conditions. The second study upgrades Harvest Commons to the same invasion logic and compares governance architectures rather than only signal choices. After a fixed-composition pilot and a lighter non-LLM invasion pass established that hybrid governance was promising, a stronger search-over-mutations stress test became the main Harvest result. In that high-power matrix, the hybrid condition wins all eight decision cells against top-down-only control, improving patch health and usually reducing neighborhood overharvest, though at a moderate average welfare cost. A narrower medium-tier PPO validation points in the same direction: hybrid self-play improves patch health and lowers local aggression and neighborhood overharvest relative to top-down-only control, with only a small welfare cost. The combined result is a two-level evaluation protocol: first identify effective central governance signals in a controlled commons, then test how broader governance architectures behave in a second substrate under the same turnover pressure.

1 Introduction

Classic commons theory predicts over-extraction under open access and highlights institutional design as the key lever for sustainable use [2, 7, 11]. In machine learning, sequential social dilemma research has made the same point in a different language: cooperation is a property of policies interacting over time in stateful environments, so evaluation must account for long horizons, partial observability, and held-out social situations rather than one-shot payoffs [1, 3–5]. Recent LLM-agent commons work shows that cooperative behavior can emerge in simulated societies, but those findings remain fragile under changes in prompts, partner populations, and environmental conditions [9, 10]. What remains weakly characterized is whether governance itself stays effective once new exploitative strategies keep entering the population.

This paper studies that problem as *governance under adversarial strategy injection*. The paper-level question is broader than a single fishery result: *how can governance stabilize cooperation under repeated strategic turnover in sequential commons, and what changes when governance moves from fixed central signals to broader governance architectures?* To answer that question cleanly, the paper is split into two linked studies. The first study uses a minimal Fishery Commons setting to identify which *central governance signals* remain effective under invasion pressure. It begins with a matched mutation-versus-live-LLM baseline, then widens into a top-down comparison over monitoring with sanctions, adaptive quotas, and temporary closures, and finally checks the resulting ranking with a learned-policy self-play validation in the same medium tier. The second study moves to Harvest Commons and asks a downstream question: once useful central signals have been identified, how should governance itself be organized? The comparison there is between purely top-down control, bottom-up local coordination, and hybrid governance, with the main claim anchored in the invasion matrix and a narrower learned-policy check used only as supporting validation.

That split is deliberate. Commons research has long shown that communication, collective choice, and locally enforced rules can materially change outcomes even without full central command [8, 12]. At the same time, benchmark work in multi-agent learning increasingly argues that robust conclusions should survive transfer across substrates and social situations, not only one carefully tuned environment [3, 5]. The Fishery Commons study therefore isolates the signal-design question, while the Harvest Commons study widens the lens to governance architecture. Together they belong in one paper because they address the same overarching problem from two adjacent levels: which central interventions work, and what changes when governance is no longer only a hard-coded central rule.

The contribution is therefore not "a new fishery environment" or "another LLM society demo." The contribution is a two-level invasion-stability protocol for sequential commons. First, we use repeated adversarial strategy turnover, held-out regime evaluation, run-level uncertainty, and a learned-policy validation to compare central governance signals in a minimal renewable commons. Second, we carry the same pressure test into a new commons substrate and compare top-down, bottom-up, and hybrid governance architectures. The resulting evidence is specific enough to support concrete claims, but broad enough to speak to decentralized AI settings where governance is part of the system design rather than an afterthought.

2 Methods

Both studies use the same core invasion protocol. A population of strategies is evaluated over generations, weaker strategies are removed, and new strategies are injected through a specified generator. Selection is performed in train conditions and reported robustness is measured in held-out test conditions. This separates adaptation pressure from generalization assessment and makes governance comparisons about out-of-distribution stability rather than only in-sample performance.

2.1 First study: Fishery Commons and central governance signals

The first study uses a minimal renewable fishery commons with a shared stock, per-agent harvest actions, regeneration, and collapse dynamics. The main invasion runs use interpretable threshold strategies, and governance changes extraction pressure directly through monitoring, quota clipping, sanctions, adaptive quotas, or temporary closures. The environment logs requested harvest, realized harvest, quota-clipped harvest, repeat-offender rate, closure activity, and stock-recovery lag so that governance can be analyzed mechanistically rather than only through final collapse outcomes.

The Fishery Commons study has two invasion layers plus one learned-policy validation. The first invasion layer is a matched baseline that compares no governance, monitoring alone, and monitoring with sanctions across two injector types: mutation and live JSON-based LLM injection through Ollama with the ‘qwen2.5:3b-instruct‘ model. This baseline is run on the easy and medium fishery tiers. The second invasion layer is a broader top-down sweep, restricted to the medium tier, that compares no governance, monitoring with sanctions, adaptive quotas, and temporary closures under three non-LLM injector families: mutation, an adversarial heuristic injector, and search over mutations. The sweep also varies partner composition between balanced and adversarial-heavy populations, and it varies adversarial pressure between 0.3 and 0.5. This second layer is what identifies which central governance signals remain effective once the setting becomes more hostile.

The learned-policy validation asks a narrower question: does the resulting governance ranking survive outside the hand-specified threshold family? For this check, agents share a parameter-shared PPO policy with a small multilayer perceptron, observe normalized stock and normalized timestep, and choose among 11 harvest bins. Training uses the same fishery train regime as the matched baseline (‘regen_rate = 2.0‘, ‘obs_noise_std = 6.0‘) for 250,000 environment steps per seed. Final checkpoint selection uses train-regime evaluation only, and held-out reporting uses the fixed ‘medium_v1‘ benchmark pack. A direct five-seed confirmatory comparison evaluates no governance against adaptive quotas, and a matched five-seed governance follow-up compares monitoring with sanctions, adaptive quotas, and temporary closures under the same PPO protocol. This is not a second invasion matrix; it is a policy-family robustness check on the top-down ranking. The shared Fishery invasion defaults for the matched baseline are 15 generations, 32 train and 32 test seeds per generation, replacement fraction 0.2, and adversarial pressure 0.3.

2.2 Second study: Harvest Commons and governance architecture

The second study introduces Harvest Commons, a second renewable commons substrate built around local patches rather than a single global stock. It now uses the same invasion logic as the first study rather than only a fixed-composition matrix. A population of six parameterized Harvest strategies is evaluated, ranked, and partially replaced over generations. The evolving strategies control local harvest fractions, restraint signals, credit requests, credit offers, reciprocity weights, and cap-compliance margins. Communication and side-payments remain condition-controlled by the environment, so the same evolving strategy family can be evaluated under different governance architectures.

The Harvest Commons study compares four governance conditions: none, top-down only, bottom-up only, and hybrid. Top-down only uses a fixed government agent that observes patch stress and local aggressive neighborhoods, then applies targeted caps. Bottom-up only enables local communication and side-payments without central intervention. Hybrid combines the fixed government agent with local communication and credit transfers. The earliest Harvest evidence is a fixed-composition pilot, which is retained only as a precursor. A later invasion pass with lighter non-LLM injectors established that hybrid governance was worth stressing further. The main Harvest evidence in the present draft then comes from a stronger search-over-mutations invader, which repeatedly proposes the most damaging child among multiple mutated candidates.

The strengthened Harvest cycle begins with a pilot matrix over the medium and hard tiers, the balanced and adversarial-heavy partner mixes, top-down only versus hybrid governance, mutation versus search over mutations, and adversarial pressures 0.3 and 0.5. The decision-critical Stage C matrix then fixes the injector to search over mutations and reruns those same medium and hard, balanced and adversarial-heavy, top-down and hybrid cells at higher power. Each Stage C cell is run for 15 generations with 32 train and 32 test seeds per generation. In addition to patch health and welfare, the Harvest Commons study reports local mechanism metrics including local aggression, neighborhood overharvest, capped-action rate, targeted-agent fraction, and prevented harvest. These are needed because a governance architecture can change *where* exploitation happens, not only the global average outcome.

The second study also includes a narrow learned-policy validation. For this check, agents share a parameter-shared PPO policy with separate communication and action heads. Communication is chosen from local observations, while harvest and credit-offer decisions condition on local observations plus inbox summaries from neighboring agents. The validation is restricted to the ‘medium_h1‘ tier and compares top-down-only against hybrid governance over five training seeds, each run for 100,000 environment steps. Training reward combines welfare with a patch-health bonus and mild penalties on local aggression and neighborhood overharvest. Held-out reporting uses the fixed ‘medium_h1‘ benchmark pack. This is a supporting architecture check, not a replacement for the main Stage C invasion matrix.

2.3 Statistical reporting

The primary matched-baseline outcome in the Fishery Commons study is held-out collapse reduction, reported as the difference between no governance and monitoring with sanctions. Supporting metrics are held-out stock, time-to-collapse, and per-regime survival. Confidence intervals for the matched baseline are run-level bootstrap intervals with 5,000 resamples. The broader Fishery Commons sweep uses the same run summaries but is interpreted as a ranking study: adaptive quota, temporary closure, and monitoring with sanctions are compared on collapse first and welfare second. The learned-policy validation reports the same held-out collapse, stock, welfare, and per-regime survival summaries over five independent PPO training seeds.

The primary comparison in the Harvest Commons study is the paired difference between hybrid and top-down-only governance at matched run identifiers within each tier, partner mix, and adversarial-pressure cell. This comparison isolates whether local communication and credit sharing add anything once central intervention is already present. Confidence intervals are again estimated by bootstrap over paired deltas. In the high-power Stage C matrix, garden failure is never the discriminating outcome, so the main Harvest Commons evidence is carried by patch health, welfare, local aggression, and neighborhood overharvest. The supporting Harvest PPO validation reports held-out means over five independent training seeds in ‘medium_h1‘; because garden failure is saturated there as well, interpretation again focuses on patch health, welfare, local aggression, and neighborhood overharvest.

Formal invasion dynamics and the saturation identifiability condition are maintained in the project notes and supplementary source files. In the empirical pipeline, the reported quantities are estimated from run-level

summaries and frozen result artifacts to avoid post-hoc threshold drift.

3 Results

3.1 First study: central governance signals in Fishery Commons

The matched Fishery Commons baseline shows that even a minimal central intervention survives repeated injection pressure. Under five paired high-power runs in the medium tier, monitoring with sanctions reduces held-out collapse relative to no governance for both injector types. For mutation, collapse reduction is 0.1099 with 95% confidence interval $[0.0310, 0.1896]$. For live JSON-based LLM injection, collapse reduction is 0.0835 with 95% confidence interval $[0.0154, 0.1509]$. Mean held-out stock also rises substantially, by 23.35 for mutation and 22.40 for live LLM injection. These matched results establish that the invasion-stability protocol is not an artifact of one injector family and that a lightweight central signal already improves robustness.

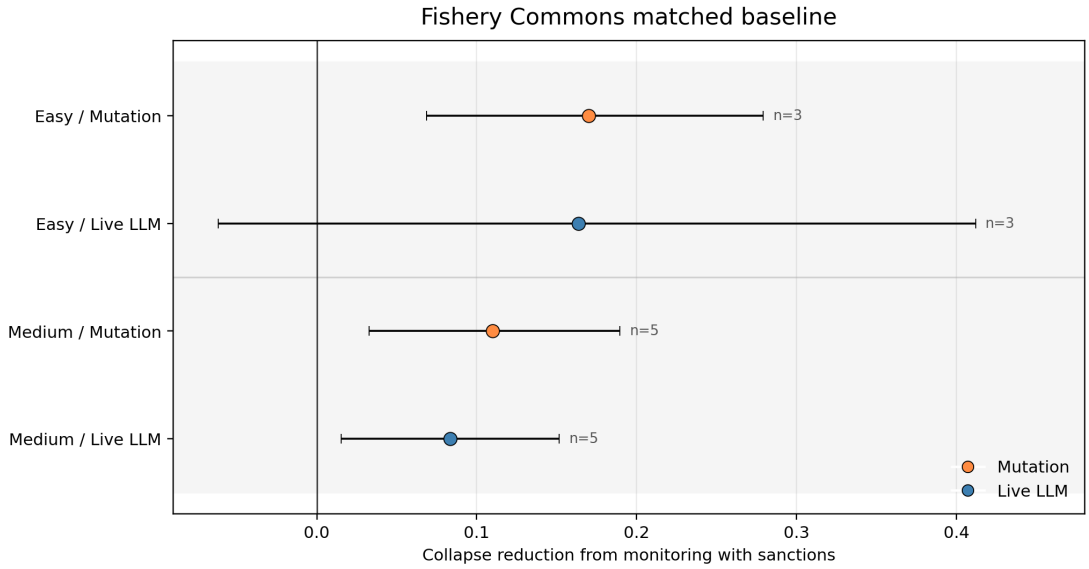

Figure 1: Fishery Commons matched baseline. Monitoring with sanctions reduces held-out collapse relative to no governance for both mutation and live LLM injection. Dots show paired-run means and whiskers show 95% confidence intervals.

The broader top-down sweep in the medium fishery tier sharpens the central-governance question. Across balanced and adversarial-heavy partner mixes, pressure levels 0.3 and 0.5, and three non-LLM injector families, adaptive quotas are the most frequent winner in every injector block. They win all four scenarios for mutation, all four for the adversarial heuristic injector, and all four for search over mutations. This happens because adaptive quotas and temporary closures tie on collapse reduction in most cells, but adaptive quotas usually retain more welfare outside the most hostile partner mixes.

Table 1: Fishery Commons medium-tier top-down comparison. Collapse reductions are measured relative to no governance, and the final column reports the welfare advantage of adaptive quotas over temporary closures.

Injector	Sanctions	Adaptive quota	Closure	Adaptive quota minus closure welfare
Mutation	0.3737	0.9167	0.9167	2408.81
Adversarial heuristic	0.0208	0.9167	0.9167	2211.61
Search over mutations	0.2617	0.9167	0.9167	2723.02

The nuance matters. In balanced partner mixes, adaptive quotas dominate closures on welfare by large margins: 4792.55 for mutation, 5202.70 for the adversarial heuristic injector, and 5367.27 for search over mutations at pressure 0.3. In adversarial-heavy partner mixes, that advantage shrinks or reverses: the same adaptive-minus-closure welfare difference is −215.62, −779.48, and −71.33 at pressure 0.3 for the three injectors. So the correct interpretation is not that temporary closures fail. It is that adaptive quotas are the strongest overall signal, while closures remain a serious fallback when ecological protection is prioritized over welfare retention.

Mechanism logging explains why monitoring with sanctions is weaker than the more forceful top-down alternatives. Mean quota-clipped harvest under monitoring with sanctions is 8.53 for mutation, 18.32 for the adversarial heuristic injector, and 9.61 for search over mutations, while mean repeat-offender rates remain 0.3032, 0.5568, and 0.3458. Monitoring with sanctions therefore reduces harm without fully suppressing aggressive extraction, whereas adaptive quotas and temporary closures push collapse to zero in nearly every cell of the first Fishery Commons study.

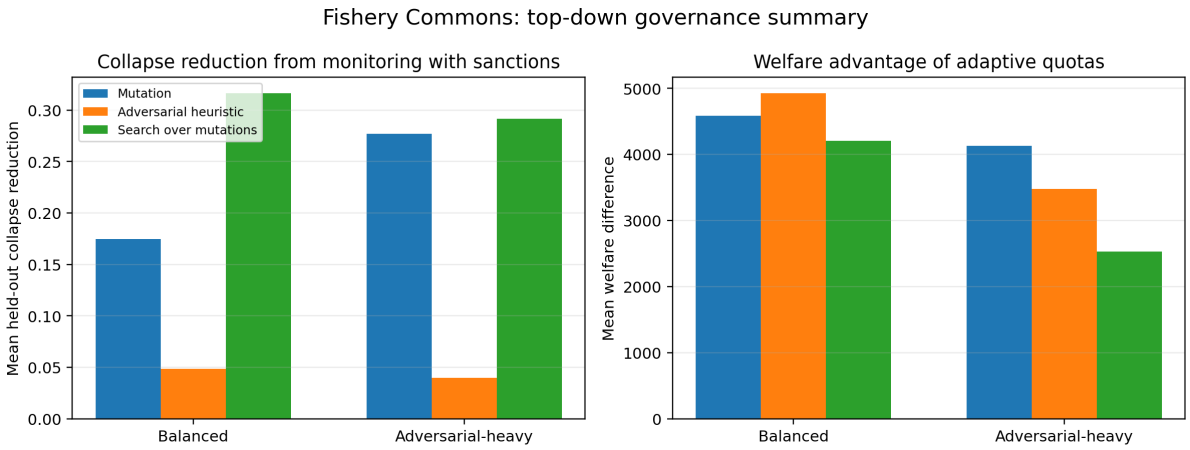

Figure 2: Fishery Commons top-down summary. Left: mean held-out collapse reduction from monitoring with sanctions relative to no governance. Right: mean welfare advantage of adaptive quotas over temporary closures. Adaptive quotas dominate in balanced settings but lose some advantage in adversarial-heavy mixes.

A learned-policy validation reaches the same conclusion. Under five PPO self-play training runs on the held-out ‘medium_v1‘ pack, adaptive quotas reduce mean collapse from 0.8141 under no governance to 0.0, raise mean stock from 97.56 to 186.05, and raise mean welfare from 4770.20 to 9902.35. A matched governance follow-up under the same PPO protocol also ranks adaptive quotas above the other governed conditions: monitoring with sanctions reaches held-out collapse 0.0016, stock 126.52, and welfare 6800.78, while temporary closures reach collapse 0.0, stock 124.05, and welfare 4719.58. So the first-study ranking is not tied only to hand-written threshold strategies. A simple learned policy family rediscovers adaptive quotas as the strongest medium-tier governance signal.

3.2 Second study: governance architecture in Harvest Commons

The second study now uses the same invasion logic as the first, rather than only a fixed-composition comparison. The earlier fixed-composition Harvest matrix and the first invasion pass with lighter non-LLM injectors both suggested that hybrid governance was worth taking seriously. The stronger test in the current draft comes from the search-over-mutations cycle. Its Stage B pilot already showed that hybrid governance survives a harsher attacker: hybrid won 13 of 16 pilot cells, while top-down-only control won the remaining 3. The main Harvest Commons evidence therefore comes from the Stage C high-power search-over-mutations matrix, which restricts attention to the decision-critical rows in the medium and hard tiers, the balanced and adversarial-heavy partner mixes, top-down-only versus hybrid governance, and pressures 0.3 and 0.5.

Table 2: Harvest Commons high-power search-over-mutations summary. Positive deltas mean hybrid governance exceeds top-down-only control; negative values are improvements for lower-is-better metrics such as local aggression and neighborhood overharvest.

Summary quantity	Value
Hybrid best-condition cells	8 / 8
Top-down-only best-condition cells	0 / 8
Mean patch-health delta	+0.0438
Mean welfare delta	-0.0918
Mean local-aggression delta	-0.0037
Mean neighborhood-overharvest delta	-0.0646

In that stronger high-power matrix, hybrid governance wins all 8 decision cells. That matters because the question is no longer whether hybrid can beat top-down-only control under a friendly attacker. The question is whether it still does so once the invader actively searches for stronger exploitative children. The answer is yes, but with a clearer trade-off than the earlier Harvest passes suggested.

The aggregated Stage C view makes that trade-off readable. Patch health rises in every single decision cell, and every patch-health confidence interval remains above zero. In the hard tier with an adversarial-heavy population, the paired patch-health delta is $+0.0157$, with a 95% confidence interval from 0.0027 to 0.0264, while the welfare delta is -0.0883, with a 95% confidence interval from -0.1668 to -0.0119. In the hard tier with a balanced population at pressure 0.3, hybrid improves patch health by $+0.1356$, with a 95% confidence interval from 0.0303 to 0.2722, but welfare falls by 0.2306, with a 95% confidence interval from -0.3922 to -0.0643.

The medium tier is more mixed, but still directionally consistent. In the adversarial-heavy block at pressure 0.3, patch health rises by $+0.0474$, with a 95% confidence interval from 0.0140 to 0.1019, while welfare overlaps a modest cost. In the balanced block at pressure 0.3, both patch health and welfare lean in hybrid's favor, though the welfare interval crosses zero. The hardest welfare penalty appears in the medium balanced block at pressure 0.5, where patch health still rises by $+0.0438$, with a 95% confidence interval from 0.0052 to 0.0863, but welfare falls by 0.2866, with a 95% confidence interval from -0.5390 to -0.0702. The second Harvest Commons study therefore does not claim that hybrid governance is free. It claims something more precise: even under a stronger search-based invader, hybrid governance preserves the commons better than top-down-only control, while the welfare price depends strongly on social composition and pressure.

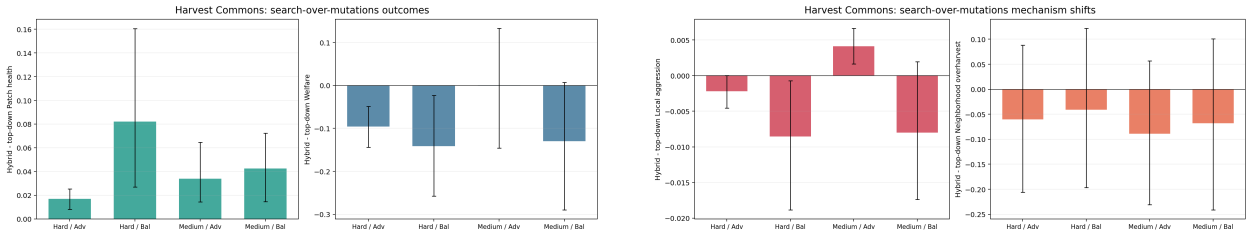

Figure 3: Harvest Commons search-over-mutations summary. Left: hybrid governance improves patch health in every block, while welfare depends on social mix and pressure. Right: negative values are improvements for lower-is-better metrics, showing that hybrid governance lowers local aggression on average and usually reduces neighborhood overharvest.

These mechanism metrics explain why the failure metric is uninformative in the Harvest Commons study. Across all 8 high-power rows, the paired garden-failure delta is exactly zero, so governance separation happens before outright garden failure. Hybrid governance matters because it shifts local behavior: averaged over the full Stage C matrix, local aggression falls by 0.0037 and neighborhood overharvest falls by 0.0646 relative to top-down-only control. The stronger result is therefore behavioral and ecological rather than catastrophic.

A narrow learned-policy validation points in the same direction. Under five PPO self-play training runs on the held-out 'medium_h1' pack, hybrid raises mean patch health from 15.5673 under top-down-only control to 15.8064, lowers mean max local aggression from 0.6 to 0.4, and lowers mean neighborhood overharvest from 9.0 to 7.65. Mean welfare falls only slightly, from 16.3757 to 16.2898. Because garden failure is again saturated at zero in this medium-tier RL setting, the learned-policy check is informative only on the same

local mechanism margins that matter in the invasion matrix. It should therefore be read as supporting evidence for the architecture ranking, not as a substitute for the stronger search-over-mutations result.

Table 3: Harvest Commons PPO validation in ‘medium_h1‘. Lower is better for local aggression and neighborhood overharvest. This is supporting evidence only; the main Study 2 claim remains anchored in the high-power search-over-mutations matrix.

Governance	Patch health	Welfare	Max local aggression	Neighborhood overharvest
Top-down only	15.5673	16.3757	0.6	9.0
Hybrid	15.8064	16.2898	0.4	7.65

Together, the two studies now give a layered invasion result. In the controlled fishery setting, central governance signals can be ranked and adaptive quotas emerge as the strongest overall top-down signal under invasion pressure. In the second substrate, once governance is allowed to become architectural rather than only signal-level, hybrid governance still produces better local ecological control than top-down-only control even under a stronger search-based invader, though the welfare effect depends on both social composition and task difficulty.

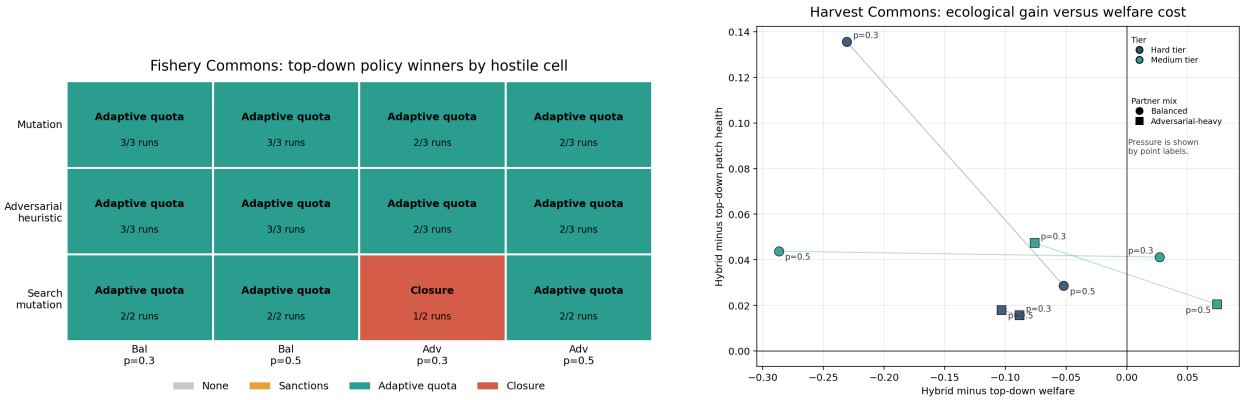

Figure 4: At-a-glance comparison of Fishery Commons and Harvest Commons. The left panel summarizes which top-down policy most often wins in the fishery setting. The right panel shows the Harvest Commons trade-off between ecological gain and welfare cost when hybrid governance is compared with top-down-only control; point color marks tier, marker shape marks partner mix, and labels indicate adversarial pressure.

4 Discussion

The paper now supports two related claims rather than one narrow claim. First, central governance signals can be ranked under adversarial strategy injection in a minimal sequential commons. The first, Fishery Commons study shows that monitoring with sanctions is already enough to improve held-out collapse under both mutation and live LLM injection, which is the minimal robustness result. The broader signal sweep then shows that stronger top-down interventions are not interchangeable: adaptive quotas are the best overall policy in medium-tier fishery runs, while temporary closures remain a credible fallback under adversarial-heavy partner mixes. A learned-policy PPO validation in the same tier reaches the same conclusion, so that ranking is not tied only to the hand-specified threshold family. This is more useful than simply saying “governance helps,” because it distinguishes light-touch enforcement from stronger adaptive intervention.

Second, governance should not be treated only as a central toggle. The second, Harvest Commons study asks what happens when the governance question is widened from signal design to architecture design. The answer is that hybrid governance can improve local ecological control beyond top-down-only control, but not for free. The earlier Harvest pilot and lighter invasion pass made that direction plausible. The stronger search-over-mutations stress test then makes it credible. In the high-power Harvest matrix, hybrid wins all 8 decision cells and improves patch health in every tier-by-partner block, but several rows now show a clearer welfare cost than before. A narrower Harvest PPO validation in the medium tier points in the same direction,

which helps show that the architecture ranking is not confined only to the hand-designed Harvest strategy family. That RL result remains intentionally secondary evidence: it is medium-tier, fixed-governance, and not itself an invasion benchmark. This is exactly the kind of trade-off that disappears if evaluation only reports a single collapse number. The Harvest results therefore justify the move from a single global stock to a second substrate with local interactions and communication.

That broader framing is why the two studies belong in one paper. The Fishery Commons study isolates which central signals work in a highly controlled environment. The Harvest Commons study then asks whether governance remains purely central once local coordination, communication, and targeted intervention are allowed to matter. Commons experiments have long suggested that communication and collective choice can improve outcomes even without a single command-and-control institution [8, 12]. The present results update that intuition for adversarially changing agent populations: once strategy turnover is explicit, governance must be evaluated both as a set of central signals and as a broader architecture for decentralized systems.

This also clarifies why people should care beyond fisheries or orchards. The contribution is not fish-specific. It is about institution design for self-interested multi-agent systems under uncertainty and turnover. The same protocol can be read as a testbed for shared compute budgets, energy systems, congestion control, spectrum allocation, or decentralized markets, where new behaviors keep entering and governance must balance resilience against performance. Recent generative-society work argues for realistic evaluation under diverse partners and conditions [6, 9, 10]; the present paper adds a more specific lesson that governance should be tested both at the level of signal choice and at the level of governance architecture, and that stronger invaders can tighten but need not overturn those conclusions.

5 Limitations

The two studies answer adjacent questions, not identical ones. The first study is a controlled signal-ranking study in Fishery Commons, while the second study is a governance-architecture extension in Harvest Commons. The second study now shares the same invasion loop and replacement logic as the first, which fixes the earlier asymmetry, but it still should not be read as a strict replication in another substrate because the governance question itself is broader.

The LLM comparison currently appears only in the Fishery Commons study as main paper evidence. This is enough to show that minimal central governance survives both mutation and live JSON-based strategy generation, but it does not yet establish the same result for Harvest Commons. A live Harvest LLM path has now been implemented and can pass a narrow reliability screen, but the matched narrow Harvest pilot still failed the manuscript's evidence gate because schema completeness remained too uneven: the mean effective-LLM fraction stayed at 0.7311 and unrepaired fallback averaged 0.2689, with missing keys as the dominant failure mode. For that reason, Harvest LLM remains capability-only evidence and the main Study 2 claim stays anchored in the stronger non-LLM search-over-mutations result.

The learned-policy extensions are still narrow. In Fishery Commons, the PPO validation is limited to the medium tier and fixed governance conditions. In Harvest Commons, a medium-tier PPO self-play validation now supports the 'hybrid ¿ top_down_only' direction, but it still uses fixed governance conditions, shaped rewards, and held-out evaluation in only one tier. These checks are enough to show that the leading Fishery and Harvest rankings are not artifacts of one hand-specified policy family, but they are not yet learned-policy invasion benchmarks and they do not yet replace the main mutation or search-over-mutations evidence.

Hard settings remain useful mainly as stress evidence. In fishery, once collapse saturates at 1.0 and survival at 0.0 across conditions, governance separation becomes non-identifiable under floor effects. In Harvest Commons, the harder tiers remain informative, but garden failure itself does not separate conditions in the high-power rows; the real signal comes from patch health, welfare, and local behavioral metrics.

Finally, both substrates remain intentionally stylized. That is a strength for identification, but it limits external realism. Market feedbacks, richer heterogeneity, richer language interaction, and explicit institutional bargaining are natural next additions once the current governance comparisons are frozen.

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